

1. (b)

$$\therefore f(x, y) \equiv x^2 + y^2 + 2gx + 2fy + c = 0 \text{ Now,}$$

$$f(0, \lambda) \equiv \lambda^2 + 2f\lambda + c = 0 \text{ and its roots are 2, 2.}$$

$$\therefore 2 + 2 = -2f, 2 \times 2 = c, \text{ i.e. } f = -2, c = 4$$

$$f(\lambda, 0) \equiv \lambda^2 + 2g\lambda + c = 0, \text{ and its roots are } \frac{4}{5}, 5.$$

$$\therefore \frac{4}{5} + 5 = -2g, \frac{4}{5} \times 5 = c, \text{ i.e., } g = \frac{-29}{10}, c = 4$$

$$= (-g, -f) = \left(\frac{29}{10}, 2 \right)$$

Hence, centre of the circle

2. (b)

The polar of the point $\left(5, -\frac{1}{2} \right)$ is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$\Rightarrow 5x - \frac{1}{2}y - 2(x + 5) + 0 + 0 = 0$$

$$\Rightarrow 3x - \frac{y}{2} - 10 = 0 \Rightarrow 6x - y - 20 = 0$$

3. (d)

$$C_1 = (-a, 0), r_1 = \sqrt{a^2 - c}; C_2 = (0, -b), r_2 = \sqrt{b^2 - c};$$

$$C_1C_2 = \sqrt{a^2 + b^2}$$

$$\therefore \text{Circles touch each other, therefore } r_1 + r_2 = C_1C_2$$

$$\Rightarrow \sqrt{a^2 - c} + \sqrt{b^2 - c} = \sqrt{a^2 + b^2}$$

$$\Rightarrow a^2b^2 - b^2c - a^2c = 0$$

$$\text{Multiplying by } \frac{1}{a^2b^2c^2}, \text{ we get } \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c}.$$

4. (d)

The equation of the common chord of the circles $x^2 + y^2 - 4x - 4y = 0$ and $x^2 + y^2 = 16$ is $x + y = 4$

which meets the circle $x^2 + y^2 = 16$ at points A(4, 0) and B(0, 4). Obviously $OA \perp OB$, where O is the origin.

Hence the common chord AB makes a right angle at the centre of the circle $x^2 + y^2 = 16$

5. (c)

The line $5x - 2y + 6 = 0$ meets the y-axis at (0, 3). We have now to find the length of the tangent from Q (0, 3) to the given circle. Hence

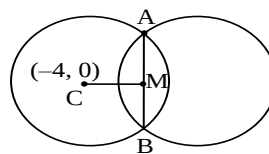
$$PQ = \sqrt{0 + 3^2 + 0 + 6 \times 3 - 2} = 5$$

6. (b)

Common chord is $S_1 - S_2 = 0$

$$4x - \mu y + 1 = 0$$

$$\therefore AC = \sqrt{15}$$



$$\frac{AB}{AM} = \frac{2}{\sqrt{6}}$$

$$\therefore CM = 3$$

$$\frac{|-16 + 1|}{\sqrt{16 + \mu^2}} = 3$$

$$\Rightarrow \mu = \pm 3$$

7. (a)

The equations of the given circles are

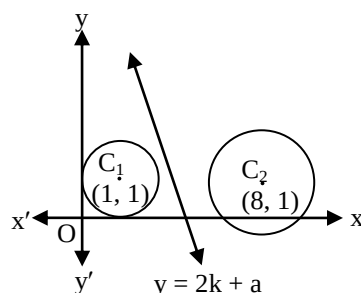
$$x^2 + y^2 - 2x - 2y + 1 = 0 \dots (1)$$

$$\text{and, } x^2 + y^2 - 16x - 2y + 61 = 0 \dots (2)$$

The coordinates of the centres and radii of these two circles are $C_1(1, 1)$, $r_1 = 1$ and $C_2(8, 1)$, $r_2 = 2$ respectively.

For the line $y = 2x + a$ not to touch or intersect circle (1), we must have

$$\left| \frac{1+a}{\sqrt{5}} \right| > 1 \quad [\text{Length of perpendicular from centre } C_1 > \text{radius } r_1]$$



$$\Rightarrow |a + 1| > \sqrt{5}$$

$$\Rightarrow a \in (-\infty, -\sqrt{5} - 1) \cup (\sqrt{5} - 1, \infty) \dots (3)$$

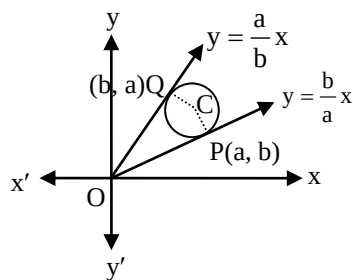
Similarly, for the line $y = 2x + a$ not to touch or intersect circle (2), we must have

$$\left| \frac{15+a}{\sqrt{5}} \right| > 2$$

$$\Rightarrow |15 + a| > 2\sqrt{5}$$

$$\Rightarrow a \in (-\infty, -15 - 2\sqrt{5}) \cup (-15 + 2\sqrt{5}, \infty) \dots (4)$$

The line $y = 2x + a$ will be between the circles, if their centres C_1 and C_2 are on the opposite sides of it.



$$\therefore (2 - 1 + a)(16 - 1 + a) < 0$$

$$\Rightarrow (a + 1)(a + 15) < 0$$

$$\Rightarrow a \in (-15, -1) \quad \dots (5)$$

From equations (3), (4) and (5), we get

$$a \in (-15 + 2\sqrt{5}, -\sqrt{5} - 1).$$

8. (a)

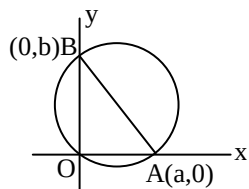
$$a + 1 = b - 1 \text{ \& } a = 2b \Rightarrow a = -4, b = -2$$

$$\Rightarrow \text{Circle } -3x^2 - 3y^2 - 6x + 9y - 3 = 0$$

or

$$x^2 + y^2 + 2x - 3y + 1 = 0$$

9. (d)



Let centroid is (α, β)

$$\alpha = a/3 \Rightarrow a = 3\alpha, \beta = b/3 \Rightarrow b = 3\beta$$

$$\text{radius} = \frac{AB}{2}$$

$$3k = \frac{\sqrt{a^2 + b^2}}{2}$$

$$\Rightarrow 6k = \sqrt{9\alpha^2 + 9\beta^2}$$

$$\alpha^2 + \beta^2 = 4k^2$$

10. (b)

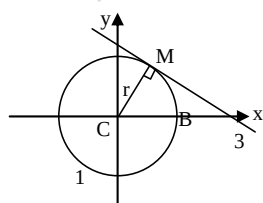
$$\text{Put } y = 0; x^2 - 4x + \lambda = 0 \text{ give equal roots so } D = 0$$

$$\Rightarrow 16 - 4\lambda = 0 \Rightarrow \lambda = 4$$

11. (c)

$$x^2 + y^2 = 9 \quad \dots (1)$$

$$3x + 4y = 0 \quad \dots (2)$$



Equation of the line parallel to the line (2) is

$$\Rightarrow 3x + 4y + \lambda = 0 \quad \dots (3)$$

$\therefore$ line (3) touches the circle (1)

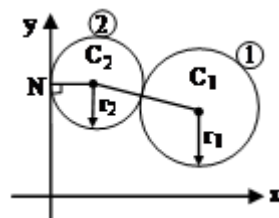
$$\Rightarrow CM = 3 \Rightarrow CM = \frac{|0 + 0 + \lambda|}{5} = 3 \Rightarrow \lambda = \pm 15$$

By (3) $\Rightarrow$ equation of tangent is $\Rightarrow 3x + 4y = 15$

12. (d)

$$x^2 + y^2 - 6x - 6y + 14 = 0 \quad \dots (1)$$

$$\Rightarrow C_1(3, 3), r_1 = 2$$



Locus of centre $C_2(h, k) = ?$

$$\therefore C_2N = r_2 = |h| \Rightarrow C_1C_2 = (r_1 + r_2)$$

$\therefore$ Both the circles touches each other externally.

13. (b)

$$S = x^2 + y^2 - 6x - 10y + p = 0$$

$$P(1, 4)$$

$$g^2 - c < 0 \Rightarrow 9 - p < 0 \Rightarrow p - 9 > 0 \quad \dots (i)$$

$$f^2 - c < 0 \Rightarrow 25 - p < 0 \Rightarrow p - 25 > 0 \quad \dots (ii)$$

$$S_1 < 0$$

$$1 + 16 - 6 - 40 + p < 0 \Rightarrow p - 29 < 0$$

$$p < 29 \quad \dots (iii)$$

comparing (i), (ii) & (iii)

$$25 < p < 29$$

14. (a)

$$x^2 + y^2 - 6x + 2y - 8 = 0 \quad \dots (i)$$

centre $(3, -1)$

Then, coordinates of the centre will satisfy the equation of the diameter.

15. (c)

Given circle is $x^2 + y^2 = 1$, $C(0,0)$ and radius = 1 and chord is $y = mx + 1$

$$\cos 45^\circ = \frac{CP}{CR}; CP = \text{Perpendicular distance from } (0, 0) \text{ to chord } y = mx + 1$$

$$CP = \frac{1}{\sqrt{m^2 + 1}} \quad (CR = \text{radius} = 1)$$

$$\cos 45^\circ = \frac{1}{\sqrt{m^2 + 1}} \Rightarrow \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{m^2 + 1}}$$

$$\Rightarrow m^2 + 1 = 2 \Rightarrow m = \pm 1.$$

